

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10 – SUB-STRAND 2.1 PROPERTIES OF WAVES: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Today you will write your Final Explanation for the driving question: Why does the matatu's horn change pitch as it passes, and why does the FM radio signal disappear inside the Kabete tunnel, and what do these two mysteries reveal about how ALL waves behave? Answer every section in the space under each prompt. Write in full sentences, use numbers and units where you can, and use evidence from the investigations, simulations and calculations we did in this unit. Draw a labelled diagram wherever a prompt asks for one. Show every step of any calculation. You may use $v = f \lambda$ and the speed of light of 3.0×10^8 m/s.

Part 1: What is a wave? The idea behind both mysteries

A student travelling from Nakuru to Nairobi hears a matatu horn and listens to Classic 105 FM on the bus radio. (a) Explain what a wave is and what it does and does not transfer. (b) Name the wave type (transverse or longitudinal) of the matatu horn sound and of the FM radio signal, and state how the particle or field vibration relates to the direction of travel in each. (c) Explain why it is useful to treat sound and radio as the same kind of thing, even though they seem so different.

(a) A wave is a disturbance that transfers energy from one point to another without transferring matter. In the PhET Sound simulation the air particles only moved back and forth around a fixed position while the disturbance travelled forward. (b) The matatu horn is a longitudinal mechanical wave: air particles vibrate parallel to the direction the sound travels, forming compressions and rarefactions between the matatu and the student's ear. The Classic 105 FM signal is a transverse electromagnetic wave: the vibration is perpendicular to the direction of travel. (c) Both carry energy from a source to a receiver and are described by the same properties: amplitude, wavelength, frequency, period and speed, linked by $v = f \lambda$. Because they share these properties, the same wave ideas can explain both the pitch change and the signal blackout.

Part 2: The matatu horn and the Doppler effect

The matatu driver holds the horn at the same setting the whole time, yet the student, in a vehicle travelling in the opposite direction, hears a

(a) This is the Doppler effect: the apparent change in frequency due to relative motion between source and observer. As the matatu approaches, it moves toward the crests it has just emitted, so successive crests are compressed into a shorter distance and more crests reach the student's ear each second. The detected frequency is higher, so the pitch is higher. As the matatu moves away, the crests are stretched out, fewer arrive per second, and the detected frequency and pitch are lower. (b) The horn's own frequency never changes; only the frequency detected by the student changes because of the relative motion. (c) Diagram: a

higher pitch as the matatu approaches and a lower pitch as it moves away. (a) Explain, using what happens to the wave crests, why the detected frequency changes. (b) Say what stays the same and what changes: the horn's own frequency or the frequency detected by the student. (c) Draw and label a diagram showing the wave crests around the moving matatu, marking the direction of the matatu's motion, and mark the positions where the student would hear a higher pitch and where a lower pitch.	matatu drawn with an arrow showing its direction of motion; circular wavefronts closely spaced in front of the matatu (the side it is moving toward), labelled 'crests compressed, higher frequency, higher pitch', and widely spaced behind it, labelled 'crests stretched, lower frequency, lower pitch'. The student hears the higher pitch while the student is in front of the matatu, before it passes (matatu approaching), and the lower pitch once the student is behind it, after it has passed (matatu moving away).
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Part 3: Wavelengths of FM and AM signals

Classic 105 broadcasts at 105.2 MHz. A typical AM station such as KBC Radio broadcasts at 567 kHz. Radio waves travel at 3.0×10^8 m/s. (a) Use $v = f \lambda$ to calculate the wavelength of each signal, showing your working. (b) Compare each wavelength with the width of the Kabete tunnel opening of about 8 m, and state which wave is smaller than the opening and which is larger. (c) A wave travels at constant speed and its frequency is doubled. Describe what happens to its wavelength and explain why.	(a) FM: $\lambda = v / f = (3.0 \times 10^8 \text{ m/s}) / (105.2 \times 10^6 \text{ Hz}) \approx 2.85 \text{ m}$. AM: $\lambda = (3.0 \times 10^8 \text{ m/s}) / (567 \times 10^3 \text{ Hz}) \approx 529 \text{ m}$. (b) The FM wavelength (about 2.85 m) is smaller than the tunnel opening (about 8 m). The AM wavelength (about 529 m) is far larger than the opening, nearly 200 times longer than the FM wavelength. (c) Speed = frequency $\times$ wavelength, so at constant speed frequency and wavelength are inversely proportional. Doubling the frequency halves the wavelength.
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Part 4: Why FM fades and AM survives in the Kabete tunnel

(a) Define diffraction and describe what you found in the	(a) Diffraction is the spreading of waves when they pass through a gap or around an obstacle. In the simulation the spreading was greatest when the gap was narrow, and a longer wavelength diffracted more through the same gap; a very wide gap gave almost
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single-slit simulation about how the spreading of waves beyond a gap changes with wavelength and with gap width. (b) Use your answers from Part 3 and your diffraction findings to explain why the FM signal fades in the Kabete tunnel while the AM signal remains audible. (c) Our explanation uses a model in which the tunnel opening acts as a gap. Suggest one limitation of this model, and say what other feature of a real tunnel might also affect radio signals. (d) Draw a labelled sketch of the wavefronts of an FM wave and an AM wave meeting the 8 m tunnel opening.	no spreading. (b) The FM wavelength (about 2.85 m) is much smaller than the tunnel opening (about 8 m), so FM waves hardly diffract; they travel nearly straight and do not bend into the shadow zone where the bus radio is. The AM wavelength (about 529 m) is far larger than the opening, so AM waves diffract strongly, bend around and into the tunnel, and the signal stays audible. (c) This is a simplified model: a real tunnel is a long structure of concrete and metal, not just a single gap, so the materials and length of the tunnel may also weaken signals, and the gap model gives us a useful explanation based on what we investigated rather than the complete picture. (d) Sketch: FM wavefronts, with closely spaced lines, pass the 8 m opening almost straight with little spreading; AM wavefronts, with very widely spaced lines, spread out almost uniformly in all directions after the opening. Each is labelled with its wavelength.
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Part 5: What the two mysteries reveal about ALL waves

Write a short, connected scientific explanation (6–8 sentences) that answers the driving question. Your explanation should: (a) explain both mysteries using wave properties; (b) show how frequency, wavelength and wave speed link the two; and (c) state one thing you thought at the start of the unit that has changed. Then choose one other wave behaviour you studied in this unit and explain whether it is the cause of the Kabete FM fade, giving a reason.	The matatu horn changes pitch because of the Doppler effect: as the matatu approaches, sound crests are compressed, so the student detects a higher frequency and a higher pitch; as it moves away, the crests are stretched and the detected frequency is lower. The horn's own frequency never changes. The FM signal from Classic 105 fades in the Kabete tunnel because its wavelength, found from $v = f \lambda$, is only about 2.85 m, much smaller than the tunnel opening of about 8 m, so it diffracts very little. The AM signal at 567 kHz has a wavelength of about 529 m, far larger than the opening, so it diffracts strongly and stays audible. Both mysteries are explained by the same properties of waves: frequency, wavelength and speed, which decide how a wave interacts with motion and with obstacles. This shows that all waves, whether sound or radio, are disturbances that transfer energy without transferring matter and obey the same rules. (Sample change in thinking: at first I thought sound and radio were unrelated and that the horn's pitch was changed by the driver; now I know the pitch change comes from relative motion.) Refraction and total internal reflection depend on a wave changing speed at a boundary between media, as in the fibre-optic TEAMS cable at Mombasa; they are not the cause of the Kabete FM fade, which our unit explains by limited diffraction.
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RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
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Understanding of wave concepts	Defines a wave as a disturbance transferring energy without matter and correctly classifies horn (longitudinal) and FM (transverse) with accurate particle motion.	Defines a wave correctly and classifies both waves, with minor gaps in describing vibration direction.	Definition is incomplete or confuses what a wave transfers, or mixes up wave types.
Doppler explanation	Explains compressed and stretched crests, relative motion, and that only the detected frequency changes, with a correct labelled diagram showing both positions.	Explains higher and lower pitch from crest spacing with a reasonable diagram but little emphasis on the unchanged source frequency.	States pitch changes without a mechanism, or suggests the horn itself changes.
Calculations with $v = f \lambda$	Correct substitution, unit conversions and answers (about 2.85 m and 529 m), with the inverse relationship explained.	Correct method with a small arithmetic or unit slip, and mostly correct reasoning about the inverse relationship.	Incorrect method or units, or no link between frequency and wavelength.
Diffraction explanation of the tunnel	Compares wavelength with the 8 m opening for both signals, explains diffraction using simulation findings, acknowledges a limitation of the model, with a clear sketch.	Explains the tunnel using wavelength compared with the opening, with a basic sketch and limited discussion of the model.	Gives a vague or incorrect cause, such as a tunnel simply blocking FM, without comparing wavelength and gap.
Synthesis and scientific communication	Connects both mysteries through shared wave properties, reflects on a change in thinking, and judges another wave behaviour against the tunnel cause using precise vocabulary.	Links both mysteries to wave properties in a clear paragraph with mostly correct terms.	Treats the two mysteries separately, with little use of wave vocabulary or evidence.

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